TradeIQ: Identifying the Best Trades for your Fantasy Team

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Widespread reliance on heuristics creates inaccuracies in player valuations

The Core Problem

40M+

People playing Fantasy Football in North America

Limited use of advanced analytics

High reliance on heuristics and “experts”

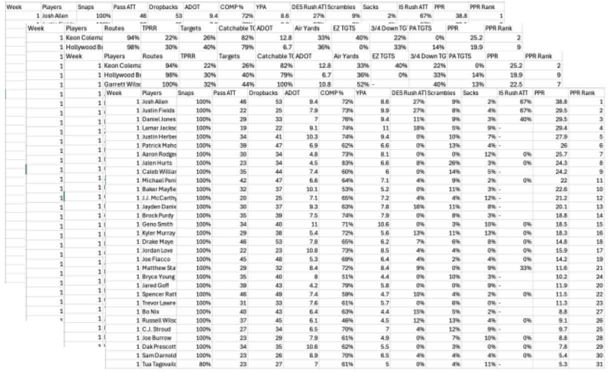
Major Opportunity for Analytics-Based Competitive Advantage

1. How can we **project future player scoring better than the field** by using advanced statistics as predictors?

2. How can we compare these projections against consensus trade values to **identify players the general public is overvaluing or undervaluing**?

Data Overview

4 Player Statistic Datasets



Contains advanced player statistics and PPR points (performance) in each of the season's 17 weeks

~6,000 Combined Rows

25 Combined Features

Pre-Week 11 Trade Value Chart

Player	Team	Trade Value
Jonathan Tay	IND	46
Christian Mc	SF	46
De'Von Achai	MIA	43
Jahmyr Gibbs	DET	41
Bijan Robins	ATL	39
James Cook	BUF	38
Josh Jacobs	GB	35
Saquon Barkl	PHI	29
Javonte Willi	DAL	27
Kyren Willian	LAR	22
Breece Hall	NYJ	22
Derrick Henr	BAL	21
Rico Dowdle	CAR	21
D'Andre Swift	CHI	21
Ashton Jeant	LV	21
Quinshon Juc	CLE	17
Chase Browr	CIN	15
Travis Etienn	JAC	14
Jaylen Warre	PIT	14
Omarion Har	LAC	13
Kimani Vidal	LAC	11
TreVeyon Hei	NE	10
Jo'quavioius I	HOU	10
Bucky Irving	TB	10
RJ Harvey	DEN	9
Aaron Jones	MIN	9
David Montgi	DET	9
Trey Benson	ARI	9
Kenneth Wal	SEA	7
Kareem Hunt	KC	7

Contains CBS Sports consensus player trade values for Week 11

~6,000 Combined Rows

25 Combined Features

Data Wrangling

1. A **participation filter** removed low-snap or low-route weeks to **prevent inaccurate data points** from injuries and garbage time

2. Week 1-10 data was **collapsed** from weekly rows into one row per player by computing **four summary features per statistic**:
   - Mean
   - Trend (slope)
   - Standard deviation
   - Last-3-week average

Methodology – Building a Predictive Model

Goal:

Project each player's aggregate points per game from weeks 11-17, using advanced statistics from weeks 1-10 as predictors

Models Used

Lasso

Selects only the most important player statistics and discards the rest, which works well when there are enough players to learn from (TE, RB, WR)

Ridge

Keeps all statistics but reduces their individual influence, preventing compressed projections resulting from having too few QBs relative to the number of features

In-Sample Results

Position	Model	N Players	Train R^2	Train MAE	Train RMSE	LOO CV MAE
TE	Lasso	101	0.649	1.81	2.44	1.97
RB	Lasso	120	0.742	2.44	3.15	2.80
WR	Lasso	182	0.650	2.35	3.06	2.87
QB	Ridge	57	0.491	4.27	5.46	4.99

Out of Sample Results

Position	N	Test MAE (pts)	Test RMSE (pts)	r
TE	17	2.93	3.97	0.677
RB	41	3.59	4.58	0.648
WR	45	3.02	3.80	0.609
QB	15	4.20	5.11	0.0079
Overall	118	3.35	4.29	0.636

Methodology – Building an Optimization-based Trade Suggester

Objective Function:

$$\max \sum_{i \in I} p_i z_i + 0.2 \sum_{i \in I} p_i (y_i - z_i)$$

*"**Maximize the starting lineup's points per game**, and add a small incentive for maximizing the bench's points per game"*

Functionality:

Generate a **set of all possible trade packages** → Optimize the model to **choose the best *m* trades**

**m = desired number of trades*

Constraints

1. **Trade Limit:** No more than 5 (*m*) trades can be made
2. **Player Trade Limit:** Players can be traded away or traded for at most once
3. **No QB Trades Allowed:** Projection accuracy wasn't good enough to trust QB trade suggestions
4. **Trade Fairness:** For each trade, consensus trade value in must be less than trade value out, but no more than 20% less
5. **Final Roster Membership:** The final roster must have 13 players, maximum 2 QBs and 2 TEs, minimum 3 RBs and 4 WRs
6. **Starter Requirement:** Starters must be on the final roster
7. **Starting Lineup Composition:** The starting lineup must be made up of 1 QB, 2 WRs, 2 RBs, 1 TE, and 1 FLEX (additional WR, RB, or TE)

Suggested Trades Showed Significant Uplift When Tested on 100 Different Lineups

Trade Suggestion Statistics

99%
Success Rate

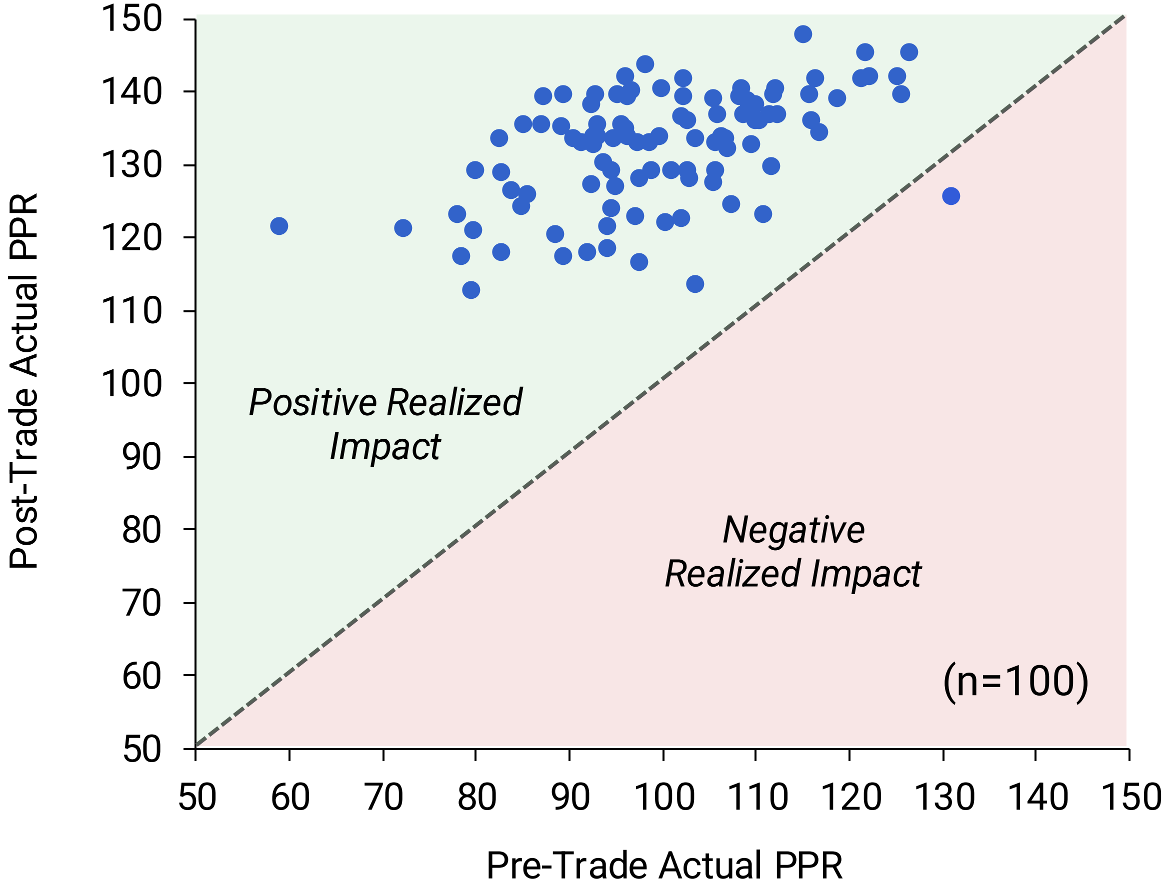
32 PPR Points
Average Uplift

What does this mean for teams?

12th Place → **3rd Place**

6th Place → **1st Place**

Realized Trade Suggestion Uplift



Insights & Limitations

Insights

1. Recent Performance Dominates
2. Market Inefficiencies Exist
3. Multi-Trade Optimization Advantage
4. Proven Real-World Impact

Limitations

1. Injuries & News Blind Spots
2. Limited Early-Season Data
3. Reliance on Historical Patterns
4. Undervalued Elite QBs & Context